

Ansh Sharma

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Education

Dr. B. R. Ambedkar National Institute of Technology Jalandhar
Bachelor of Technology (B.Tech)

July 2022 – June 2026

Experience

Siemens – Technical Intern (Software & Logic Automation)

June 2025 – August 2025

- **Logic Optimization:** Engineered high-level **State Machine logic** for automated production systems, reducing **computational overhead** and enhancing operational reliability by **15%**.
- **System Orchestration:** Implemented **modular low-latency logic frameworks** to ensure **fault-tolerant performance**.
- **Quality Assurance:** Directed **unit testing** and **algorithmic debugging** to identify critical system bottlenecks.

Projects

LinkedOut — Agentic AI & Distributed Automation Engine

FastAPI, LangGraph, AWS (EC2/S3), PostgreSQL (pgvector), Redis, Playwright, Docker

[GitHub](#) | [Live Demo](#)

- **Architected a Cloud-Native Multi-Agent System:** Engineered a stateful **LangGraph workflow** deployed on **AWS EC2**; implemented **MemorySaver checkpointing** to persist agent state across **async cycles**, reducing manual networking time by **97%**.
- **Engineered Scalable Storage & Stealth Automation:** Integrated **AWS S3** for resilient document storage and developed a **Playwright engine** with **randomized heuristics** to bypass anti-bot detection with **100% reliability**.
- **High-Precision RAG & Semantic Search:** Built a context-aware outreach engine using **pgvector** for **cosine similarity**; dynamically retrieves relevant resume sections to generate **hyper-personalized** connection notes.
- **Optimized Backend Performance:** Designed a **non-blocking FastAPI** architecture using **ThreadPoolExecutor** to decouple browser automation; utilized **Redis-backed WebSockets** for **real-time terminal log streaming**.

Autonomous Agentic RAG Engine — Generative AI & NLP Pipeline

FastAPI, React, Groq API, Llama 3.2, LangGraph, FAISS

[GitHub](#) | [Live Demo](#)

- Developed a **production-ready RAG pipeline** to query 500+ pages of documentation with **sub-second response times** using **Groq API** and **FAISS vector database**.
- Implemented **Recursive Character Splitting** and **Semantic Chunking**, increasing context retrieval precision by **30%**.
- Leveraged **LangGraph nodes** for autonomous **”Self-Correction” loops** to minimize **hallucination rates**.

AI Food Nutrition Analyzer — Full-Stack AI Web Application

React, Python, Flask, PyTorch, Edamam API, Recharts

[GitHub](#)

- Engineered a **Flask backend** to serve a custom-trained **PyTorch (EfficientNet)** model for real-time **image classification**.
- Implemented a hybrid data pipeline with **JSON-based caching** to minimize **API latency** and external calls.

Technologies

Languages: C++, SQL, Python, JavaScript

Frameworks & Libraries: FastAPI, React, Next.js, LangGraph, LangChain, Redis, Flask, Tailwind CSS

Databases: PostgreSQL, pgvector, FAISS (Vector DB)

Cloud & DevOps: AWS (EC2, S3), Docker, Nginx, GitHub Actions (CI/CD)

Tools & Platforms: Playwright, Git, Groq API, Postman, Hugging Face, Jupyter

Achievements

- Achieved a **global rank of 4,000** in **LeetCode Biweekly Contest**.
- Solved **200+ problems on LeetCode**, **100+ on Coding Ninjas**, and **100+ on GeeksforGeeks**.
- **Core Member**, LADC Club (Debating Society).